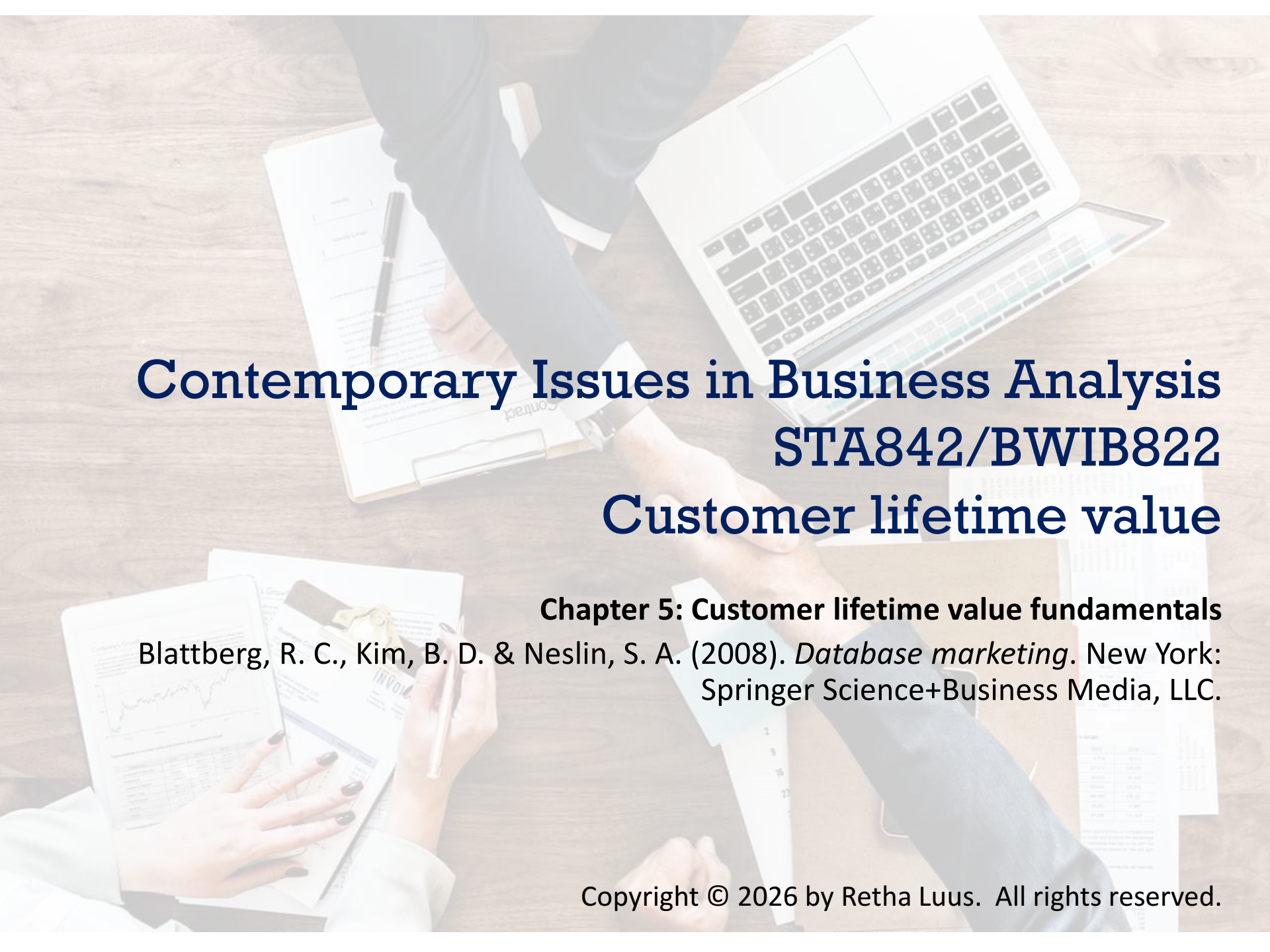


A top-down view of a business meeting around a wooden table. Several people's hands and arms are visible, interacting with documents, a laptop, and a tablet. One person is pointing at a document, another is using a pen, and a third is holding a tablet displaying a line graph. The scene is professional and collaborative.

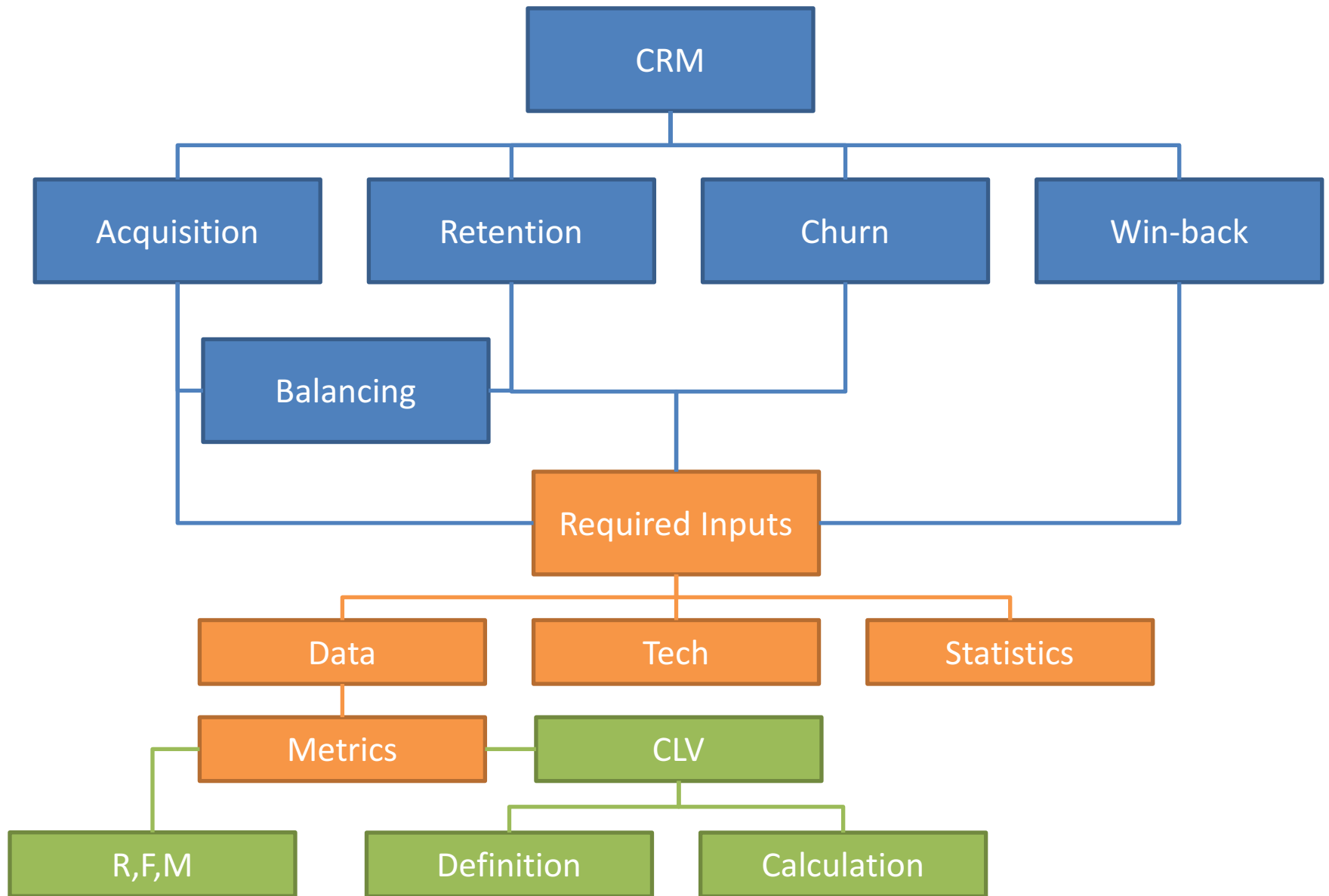
Contemporary Issues in Business Analysis

STA842/BWIB822

Customer lifetime value

Chapter 5: Customer lifetime value fundamentals

Blattberg, R. C., Kim, B. D. & Neslin, S. A. (2008). *Database marketing*. New York: Springer Science+Business Media, LLC.



Definition

- Customer lifetime value (**CLV** or **LTV**)
 - Used to be **new**, now well-known in marketing departments
 - **Cornerstone** of database marketing
 - **Quantifies** customer's **long-term value** to the firm
- Two main **applications**
 1. **Diagnose health of business**
 - Provides long-run economic view
 - Generates diagnostics based on
 - Retention rates
 - Sales per customer
 - Costs
 - **Parameters** that determine it

Definition

- Two main **applications** (cont.)
 2. Assists with making **tactical decisions**
 - Determine how much can be invested in customer acquisition
 - Decide how much to offer existing customers
- Formal **definition** of CLV/LTV

Kumar and Petersen (already familiar):

Net economic value of a customer to a firm over their entire lifetime with the firm

Blattberg et al. (today):

The net present value of the profits linked to a specific customer, once the customer has been acquired, after subtracting incremental costs associated with marketing, selling, production and servicing over the customer's lifetime

Definition

- As implied by the definition, to calculate CLV, a **firm needs to**:
 - Forecast **future sales** of a customer
 - Compute **incremental costs** per customer
 - Determine relevant **discount rate**
- Acquisition cost not included in LTV (*...once the customer has been acquired...*)
- Instead, **display acquisition cost alongside CLV/LTV**:

$$\text{Customer Equity} = \text{LTV} - \text{Acquisition Cost}$$

Definition

Simple **example** to illustrate CLV calculation over a 10-year period for some customer

Blattberg et al. (2008), p. 106

PARAMETERS (per customer)	
Retention rate	80%
Revenues if still a customer	\$200.00
Profit margin	50%
Gross profit if still a customer	\$100.00
Marketing cost if still a customer	\$20.00
Net annual profit if still a customer	\$80.00
Discount rate	15%

LTV calculation:

Year	Survival rate	Expected profit	Discount multiplier	Net discounted profit	Cumulative profit
1	1.000	\$80	1.000	\$80	\$80
2	0.800	\$64	0.870	\$56	\$136
3	0.640	\$51	0.756	\$39	\$174
4	0.512	\$41	0.658	\$27	\$201
5	0.410	\$33	0.572	\$19	\$220
6	0.328	\$26	0.497	\$13	\$233
7	0.262	\$21	0.432	\$9	\$242
8	0.210	\$17	0.376	\$6	\$248
9	0.168	\$13	0.327	\$4	\$253
10	0.134	\$11	0.284	\$3	\$256
				\$256	
				LTV	

- **Survival rate** in year t

$$0.8^{t-1}$$
- **Discount multiplier** in year t

$$\frac{1}{(1 + d)^{t-1}}$$
- **Acquisition cost** per customer:

Mail cost per prospect	\$2.00
Nr of prospect mailings	1000000
Total mail costs	\$2 000 000.00
Response rate	1%
Nr of customers acquired	10000
Cost per acquired customer	\$200.00

➤ $LTV > Acquisition\ cost\ (AC)$



➤ **Note:** Break-even in Year 4
 $(AC \approx LTV)$

Definition

Simple CLV model:

$$LTV = \sum_{t=1}^{\infty} \frac{E(\tilde{V}_t)}{(1 + \delta)^{t-1}}$$

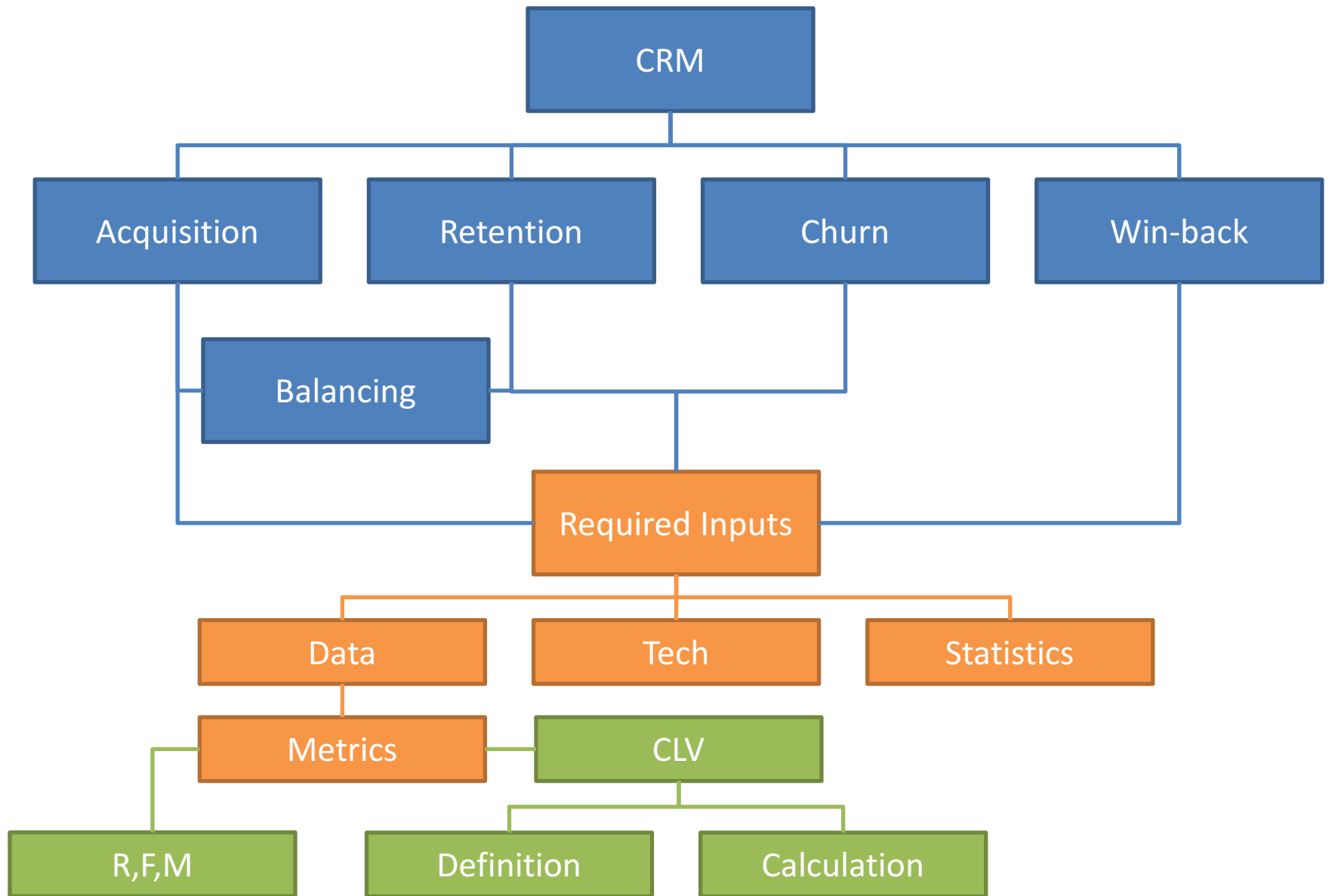
- $\tilde{V}_t$: random variable, represents customer's net profit contribution during time t
 - $E(\tilde{V}_t)$: expected net profit contribution (will need model to predict)
- δ : discount rate
- **Ideally** calculated at **customer level**, **often** on **“average” customer** due to data limitations
- Calculate **individual LTVs** and **then average** to group level (heterogeneous retention rate per customer/segment)
 - $E[f(X)] \neq f[E(X)]$

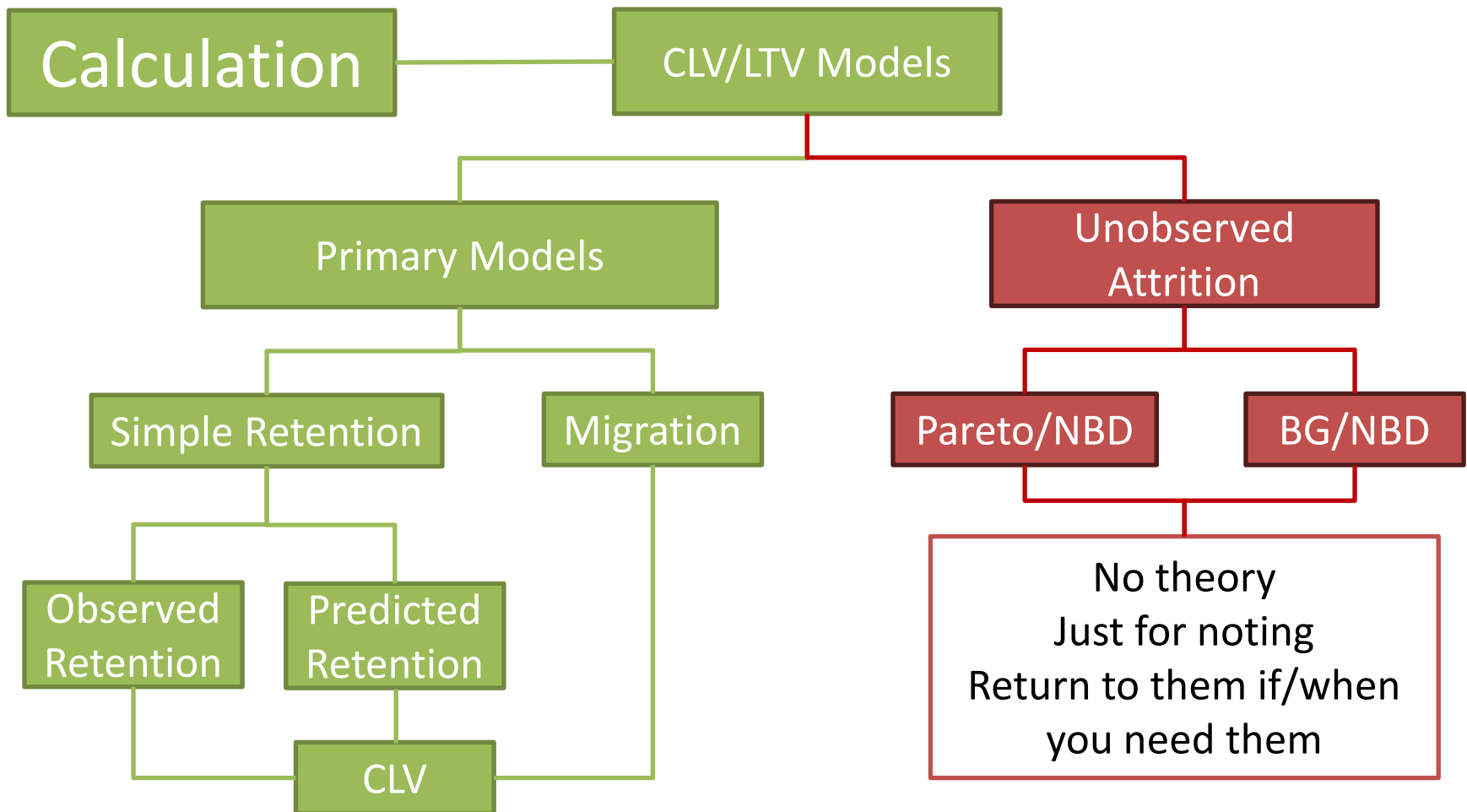
Definition

- Decomposition of expected net profit $E(\tilde{V}_t)$
$$E(\tilde{V}_t) = E(\tilde{R}_t - \tilde{C}_t) = E(\tilde{R}_t) - E(\tilde{C}_t)$$
 - $\tilde{R}_t$: revenue generated in period t
 - $\tilde{C}_t$: costs of goods, marketing and servicing
- Assumption**: future **costs** are **known**, **revenues** are **random**
- To calculate/estimate **expected lifetime value**, we'll need to calculate/estimate **expected revenue**:

$$\begin{aligned} E(\tilde{R}_t) &= P(\text{Survived until period } t) \cdot E(\tilde{R}_t | \text{Survive until period } t) \\ &= S_t \cdot E(\tilde{D}_t) \end{aligned}$$

- S_t : estimate with **hazard model**
- $E(\tilde{D}_t)$: estimate with **regression model**





Calculation

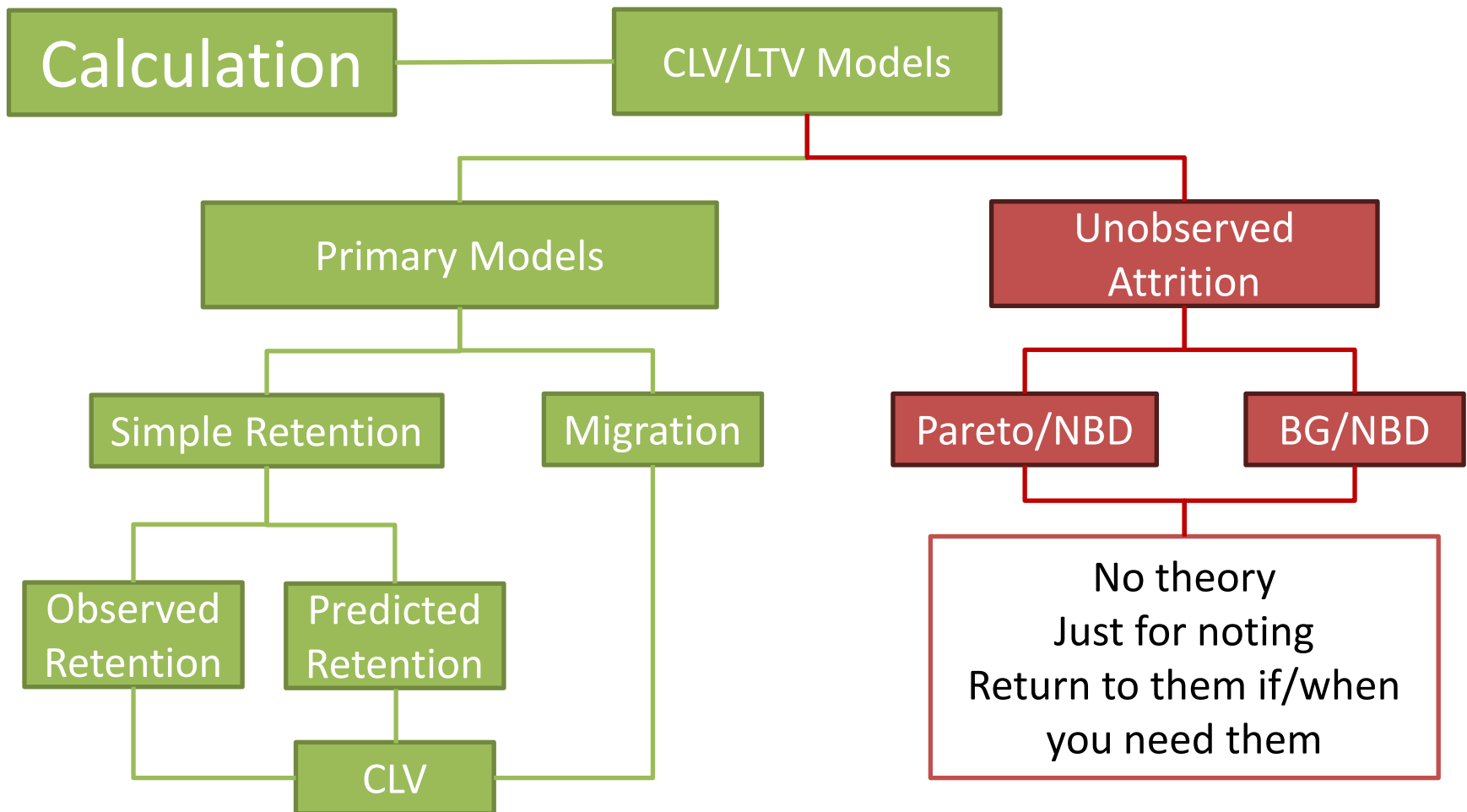
Primary CLV Models

Simple Retention

Assumption: once customer has churned/attired, customer is lost to company
Applicable: financial services, B2B, magazine subscription

Migration

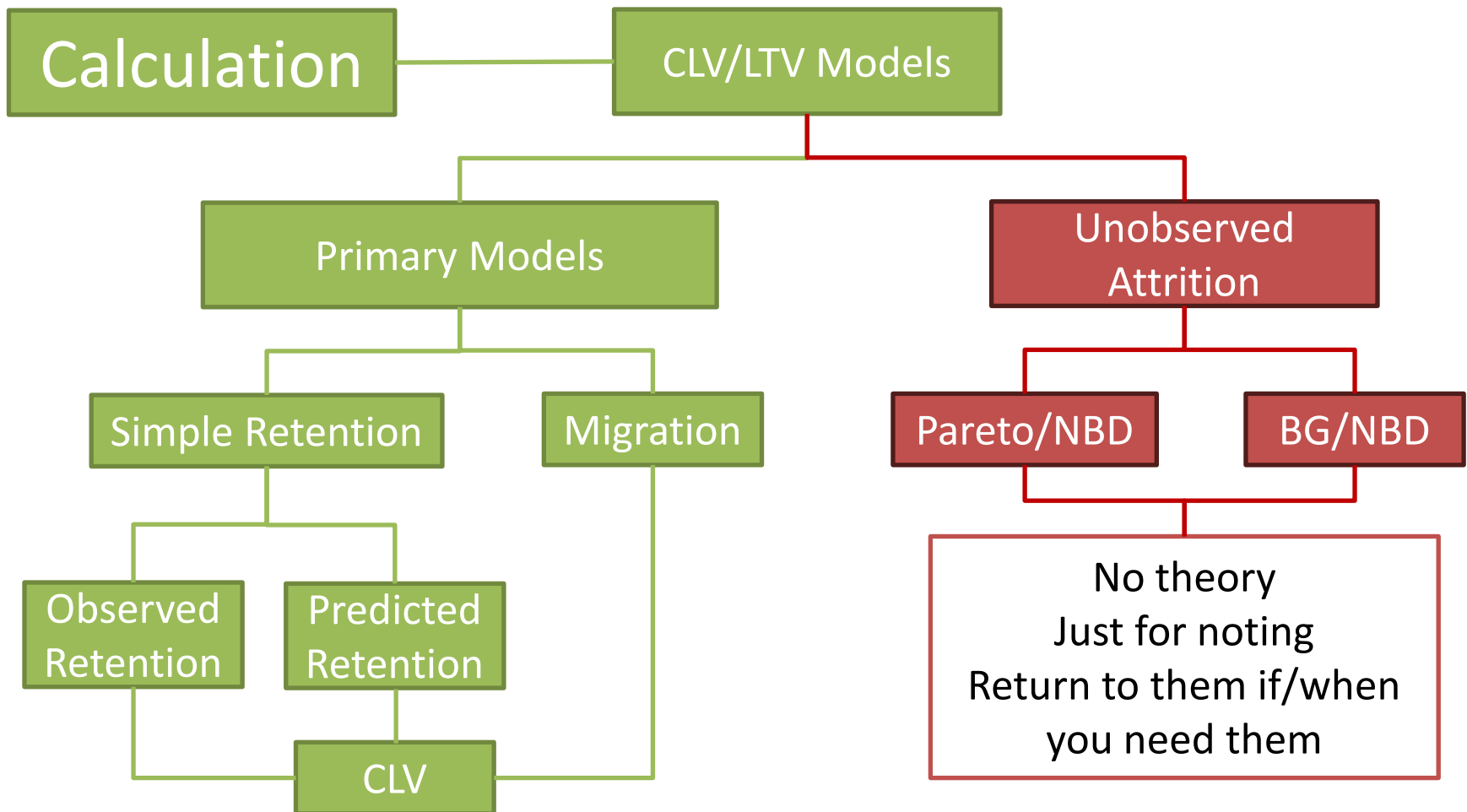
Assumption: customers can migrate between states (bought this month, last month, etc.)
Applicable: retail, catalogs



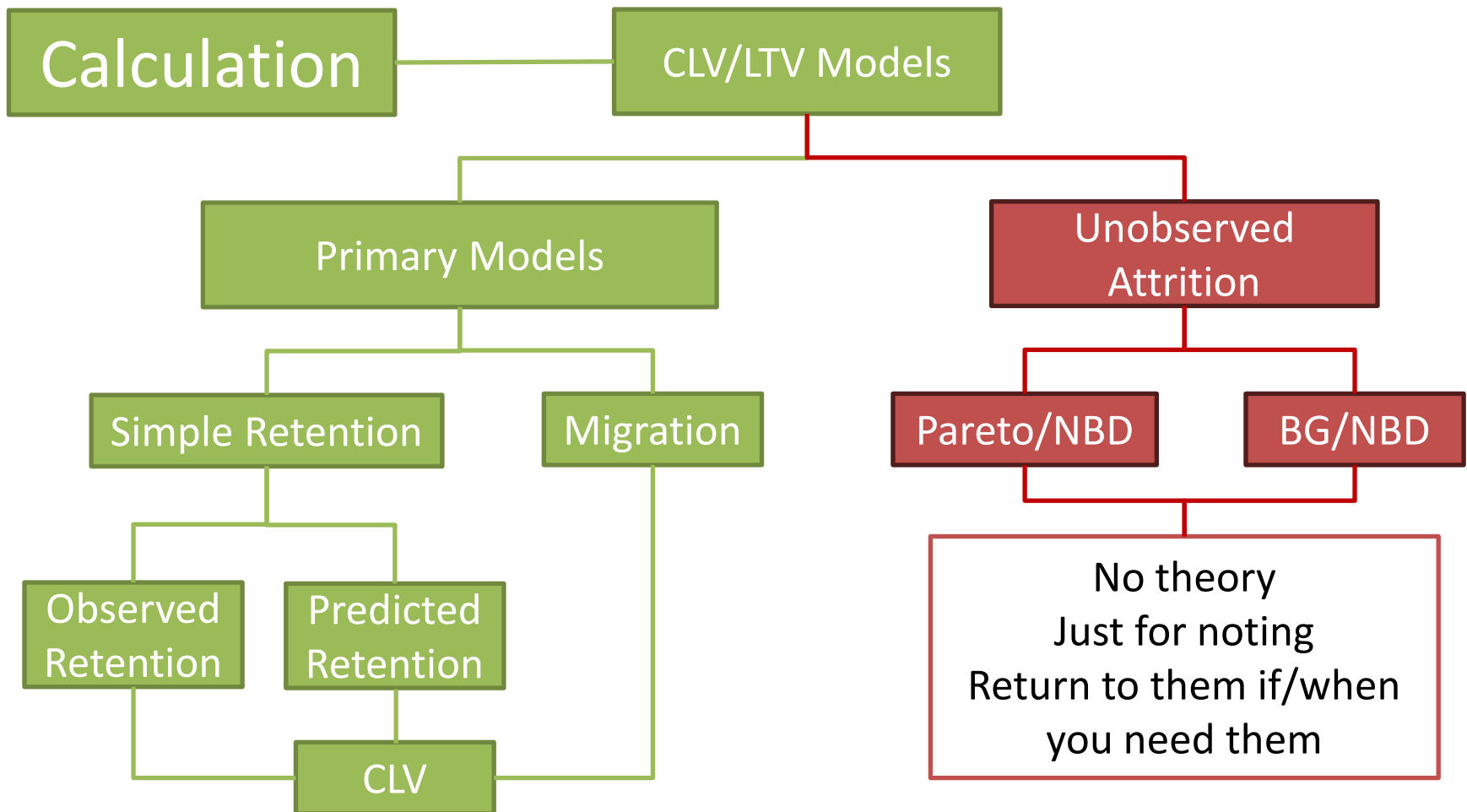
Calculation

Observed Retention

- Retention rate **one of most important parameters** of simple retention model
 - **Also important**: discount rate
 - Calculate retention rate by **direct observation** (simplest way)
 - Use customer base for year 1
 - Calculate % of customers that remained with company in year 2
 - **Issues**
 - Retention rate assumed to apply for **all periods** (past → future)
 - **Little flexibility** to calculate probability customer still a customer 1 or 2 or 12 years from now
 - **Little flexibility** to calculate customer-level retention rates
- **Improvement**: using **hazard models** predict the retention rates



- **Recall**: retention rate $r = 1 - c$, where c is the churn rate
- **Hazard models** (Acquisition Part 3) used to estimate **hazard rate** – instantaneous rate of event occurrence at time t
 - **Event**: churn
 - Hazard rate represents **churn rate**
 - **Cox PH model predicted hazard rates can be converted to retention rates!**
- Also, this is a model that can **incorporate customer-level covariates**
 - Obtain **per-customer retention rate!**
- **Note**: only applicable in contractual settings (observed churn)



Calculation

Simple Retention

Observed
Retention

Predicted
Retention

CLV?

- Now, retention rate $r = 1 - h$ (h : hazard rate – slide 15)
 - Survival rate in period t : r^{t-1}
- Revenue in period t : R_t (slide 8)
- Cost in period t : C_t (similar approach as for Revenue)
- Value of a customer (CLV) up to period τ (finite period):

$$LTV_{\tau} = \sum_{t=1}^{\tau} \frac{r^{t-1} \cdot (R_t - C_t)}{(1 + \delta)^{t-1}} \neq \boxed{LTV_{\infty}} \text{ ?}$$

Calculation

Simple Retention

Observed
Retention

Predicted
Retention

CLV?

- Value of a customer up to infinite time period:

$$LTV_{\infty} = \sum_{t=1}^{\infty} \frac{(R_t - C_t) \cdot r^{t-1}}{(1 + \delta)^{t-1}} = \sum_{t=1}^{\infty} (R_t - C_t) \left[\frac{r}{1 + \delta} \right]^{t-1}$$

- Infinite geometric series: $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$

$$\blacktriangleright a = (R - C), r = \left(\frac{r}{1+\delta} \right) \Rightarrow LTV_{\infty} = \frac{(R-C)(1+\delta)}{(1+\delta-r)}$$

Blattberg et al. Table 5.3

Table 5.3, **example 1**, p. 110

PARAMETERS	
Hazard rate	0.1
Revenue	\$200.00
Cost	\$100.00
Discount rate	10%

$\Rightarrow r = 0.9$

$\Rightarrow \delta$

Year	Hazard	Retention rate	Survival rate	Discount multiplier	Discounted expected profit
1	0	1	1.00	1.00	\$100.00
2	0.1	0.9	0.90	0.91	\$81.82
3	0.1	0.9	0.81	0.83	\$66.94
4	0.1	0.9	0.73	0.75	\$54.77
5	0.1	0.9	0.66	0.68	\$44.81
6	0.1	0.9	0.59	0.62	\$36.66
7	0.1	0.9	0.53	0.56	\$30.00
8	0.1	0.9	0.48	0.51	\$24.54
9	0.1	0.9	0.43	0.47	\$20.08
10	0.1	0.9	0.39	0.42	\$16.43
➤ Survival rate: r^{t-1}					\$476.06
(slide 17)					LTV₁₀
					\$550.00
					LTV_∞

- **Assume:** survival rate follows geometric distribution

- Discrete exponential
- Hazard stays constant
- $r = 1 - h \forall t$

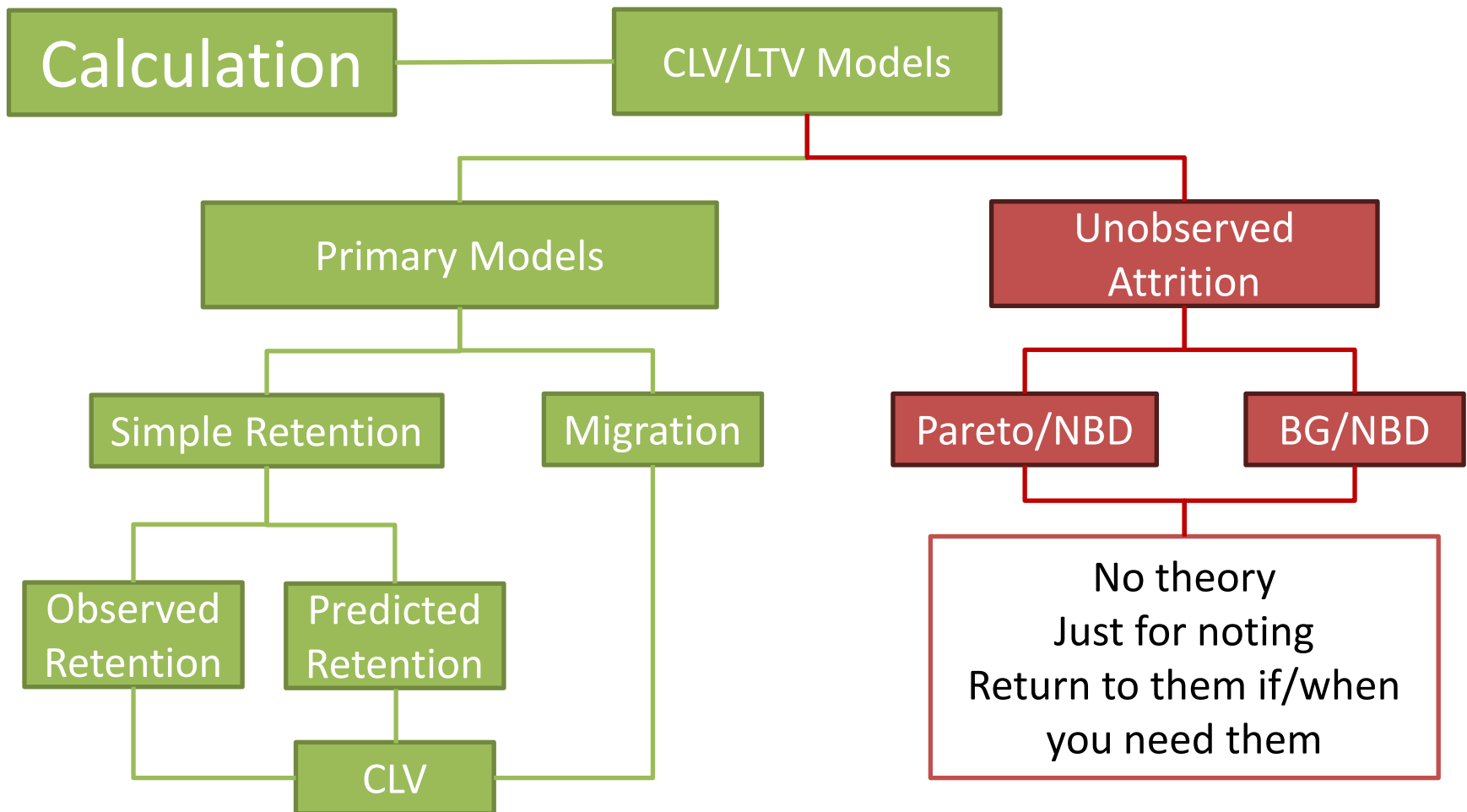
- LTV after **10 years**

$$LTV_{10} = \sum_{t=1}^{10} \frac{(R_t - C_t) \cdot r^{t-1}}{(1 + \delta)^{t-1}}$$

- LTV over **infinite horizon**

$$LTV_{\infty} = \frac{(R - C)(1 + \delta)}{1 + \delta - r}$$

➤ Try Table 5.3 example 2!!



Calculation

Migration

- **Applicable**: retail, catalogs
- Customers can **migrate between states** (bought this month, last month, etc.)
- Common definition of **states** is in terms of **recency**
 - Acknowledges that **customers can skip periods without churning**
 - Simple retention models do not allow for this
- States can be **non-absorbing** or **absorbing**
- Key **parameters**:
 - Recency/transition probabilities (**transition matrix**)
 - Profit vector
 - Discount rate
 - **Combined** they give the LTV of customers in different recency states

Calculation

Migration

- **Assumptions**

- After successful acquisition of the customer, the firm expects to receive a net contribution ($\$MC$) on the customer's initial purchase and on each succeeding purchase
- Purchases are made at most **once a period** (end of period)
- Periods are of **equal length**
- Per-period **discount rate** is d
- **Probability of purchase** in any period is a function of the customer's recency, j , and is denoted by $p_j, j = 1, \dots, NR$
- Sometimes assumed that when customer reaches recency state NR , customer is lost forever (**absorbing state**)
- In each period the firm **spends** $\$M$ **on retaining** the customer

Calculation

Migration

- Assume **3 recency states**: R_1, R_2, R_3 ($\Rightarrow NR = 3$)
 - R_1 : bought last period
 - R_2 : bought two periods ago
 - R_3 : bought ≥ 3 periods ago (**non-absorbing**)
- Table with **probability of purchasing** in each recency state

Recency state j	1	2	3
p_j	p_1	p_2	p_3

- Next, create the **transition matrix**

- The transition probability matrix P :

$$P = \begin{array}{c} \text{Period } t \\ \begin{array}{c} R_1 \\ R_2 \\ R_3 \end{array} \end{array} \begin{array}{c} \text{Period } t + 1 \\ \begin{array}{ccc} R_1 & R_2 & R_3 \end{array} \end{array} \left[\begin{array}{ccc} p_1 & 1 - p_1 & 0 \\ p_2 & 0 & 1 - p_2 \\ p_3 & 0 & 1 - p_3 \end{array} \right]$$

- Elements of P : probability customer migrates from one state to another
 - Customer in R_1 : will buy in current period with probability p_1 and remain in R_1 ; will not buy in current period with probability $1 - p_1$, moves to R_2
 - Customer in R_2 : will buy in current period with probability p_2 and move to R_1 ; will not buy in current period with probability $1 - p_2$, moves to R_3 (R_2 to R_2 has 0 probability)

- One-step transition matrix

Calculation

Migration

- **t-step** transition matrix:

$$P^t$$

- E.g., 3-step: $P^3 = P \times P \times P$: **3 periods from now (t)**
 - p_{ij} : probability customer that starts in state i will be in state j after t periods
 - p_{23}^3 : probability customer that starts in state 2 will be in state 3 after three periods

- **Assume** firm:
 - **Earns** $\$MC$ if customer purchases
 - **Spends** $\$M$ on customer to repurchase
- Define the **profit vector** $\mathbf{G}$ as (also payoff vector):

$$\mathbf{G} = \begin{bmatrix} MC - M \\ -M \\ -M \end{bmatrix}$$

- Row 1: profit contribution if customer is in R_1 (just purchased)
- Rows 2 & 3: profit contribution if customer is in R_2 or R_3 (didn't purchase, but spent $\$M$ on marketing)

Calculation

Migration

- Vector of expected profit after one period:

$$\Pi_1 = PG$$

- Vector of expected profit after two periods:

$$\Pi_2 = P^2 G$$

⋮

- Vector of expected profit after T periods:

$$\Pi_T = P^T G$$

- Vector of expected net present value after one period:

$$\frac{PG}{(1+d)}$$

- Vector of expected net present value after two periods:

$$\frac{P^2 G}{(1+d)^2}$$

⋮

- Vector of expected net present value after T periods:

$$\frac{P^T G}{(1+d)^T}$$

- Vector of **total net present value** from period 0 to T :

$$\mathbf{V}^T = \sum_{t=0}^T \frac{\mathbf{P}^t \mathbf{G}}{(1+d)^t}$$

- Elements of $\mathbf{V}^T$ give CLV/LTV after T periods for each customer that starts in each of the different recency states
 - 1st element of $\mathbf{V}^T$: LTV of customer that starts in R_1 , i.e., LTV of just-acquired customer, after T periods
 - 2nd element of $\mathbf{V}^T$: LTV of customer currently in R_2 , i.e., purchased two periods ago, after T periods
 - Etc.

Calculation

Migration

- Vector of **total net present value** for infinite time horizon ($T \rightarrow \infty$):

$$\mathbf{V}^{\infty} = \lim_{T \rightarrow \infty} \mathbf{V}^T = [\mathbf{I} - (1 + d)^{-1} \mathbf{P}]^{-1} \mathbf{G}$$

- $\mathbf{I}$: identity matrix
- Elements of $\mathbf{V}^{\infty}$ give long-term CLV/LTV for each customer that starts in each of the different recency states
 - 1st element of $\mathbf{V}^{\infty}$: LTV of customer that starts in R_1 , i.e., LTV of just-acquired customer
 - 2nd element of $\mathbf{V}^{\infty}$: LTV of customer currently in R_2 , i.e., purchased two periods ago
 - Etc.

Calculation

Migration

Example

- Suppose there are **three recency states** plus a fourth state which is **absorbing**, i.e., the customer is regarded as a former customer when they enter this state. The recency state table is as follows:

State (j)	1	2	3	≥ 4
p_j	0.5	0.2	0.1	0

- The net contribution when a customer purchases is \$100
- The marketing expense is \$0
- The discount rate is 10%

Calculation

Migration

Example

State (j)	1	2	3	≥ 4
p_j	0.5	0.2	0.1	0

➤ One-step transition matrix $\mathbf{P}$:

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.2 & 0 & 0.8 & 0 \\ 0.1 & 0 & 0 & 0.9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Calculation

Migration

Example

- The net contribution when a customer purchases is \$100
- The marketing expense is \$0
- The discount rate is 10%
- The profit vector $\underline{G}$ (or payoff vector):

$$\underline{G} = \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 100 - 0 = 100 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Calculation

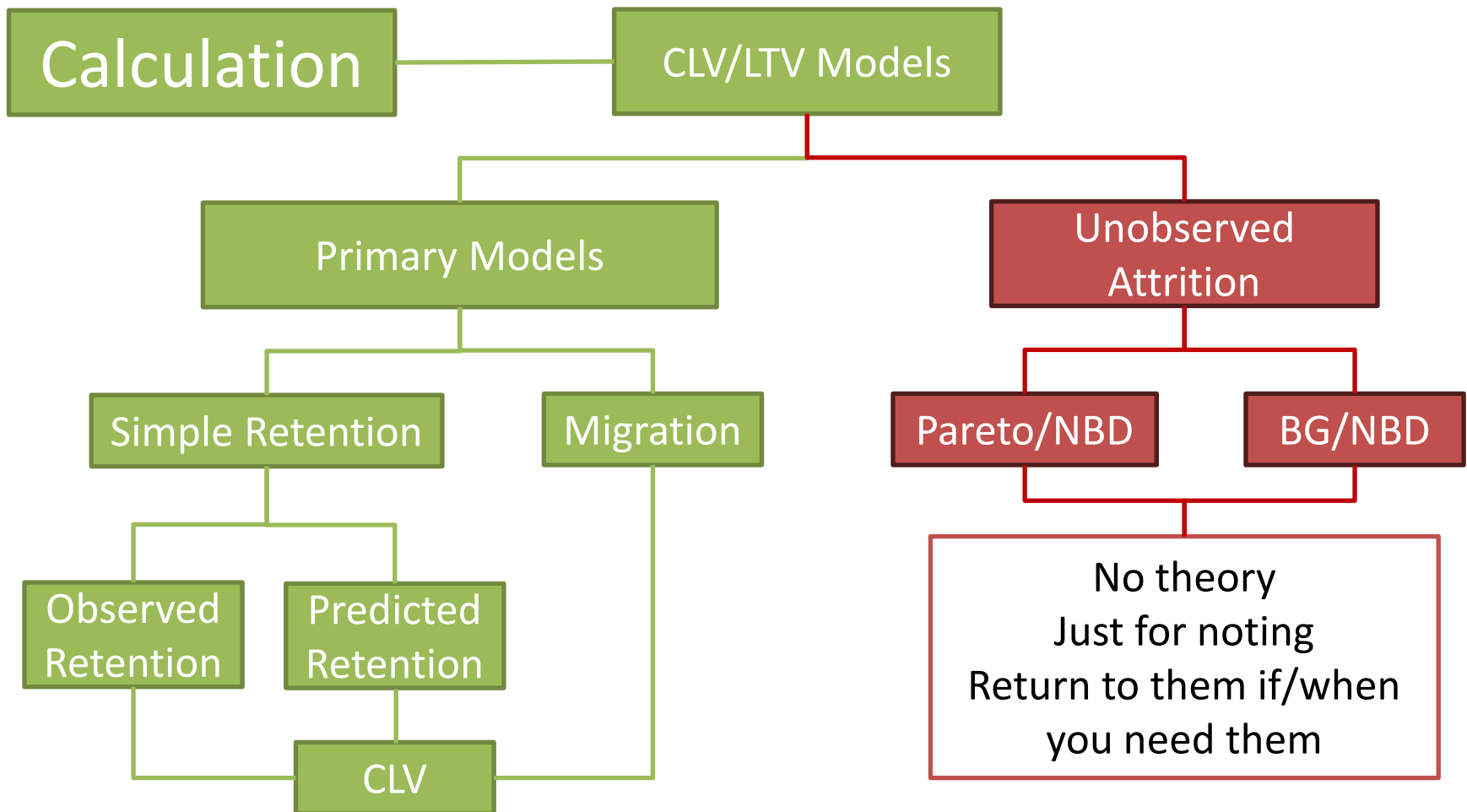
Migration

Example

1. Calculate $\mathbf{P}^2$ and $\mathbf{P}^4$
2. What is the probability of a customer starting in state 2 to end up in state 3 after 4 time periods?
3. Calculate $\Pi_5 = \mathbf{P}^5 \mathbf{G}$
4. If the firm has a 5-period view on determining which customer it should market its products to, is it profitable to market to customers in every state? Explain.
5. Calculate V^{15} and V^∞



Work through other examples in Blattberg et al. (2008), p. 115-121



Calculation

CLV/LTV Models

Unobserved Attrition

Fundamental question: is the customer “alive” or “dead”?

Attrition easy to observe in contractual settings

- Telecom, magazines, network, etc.
- Customer doesn't renew contract

Attrition not observed in non-contractual settings

- Travel services, retail, restaurants, etc.
- Wasting marketing budget sending catalogs to customers that have attrited

Models developed to date focus on four key phenomena:

- The number of purchases in a given time period
- Heterogeneity in the parameters of the purchase rate model
- Customer lifetime, i.e. how long the customer is alive
- Heterogeneity in the parameters governing the lifetime model

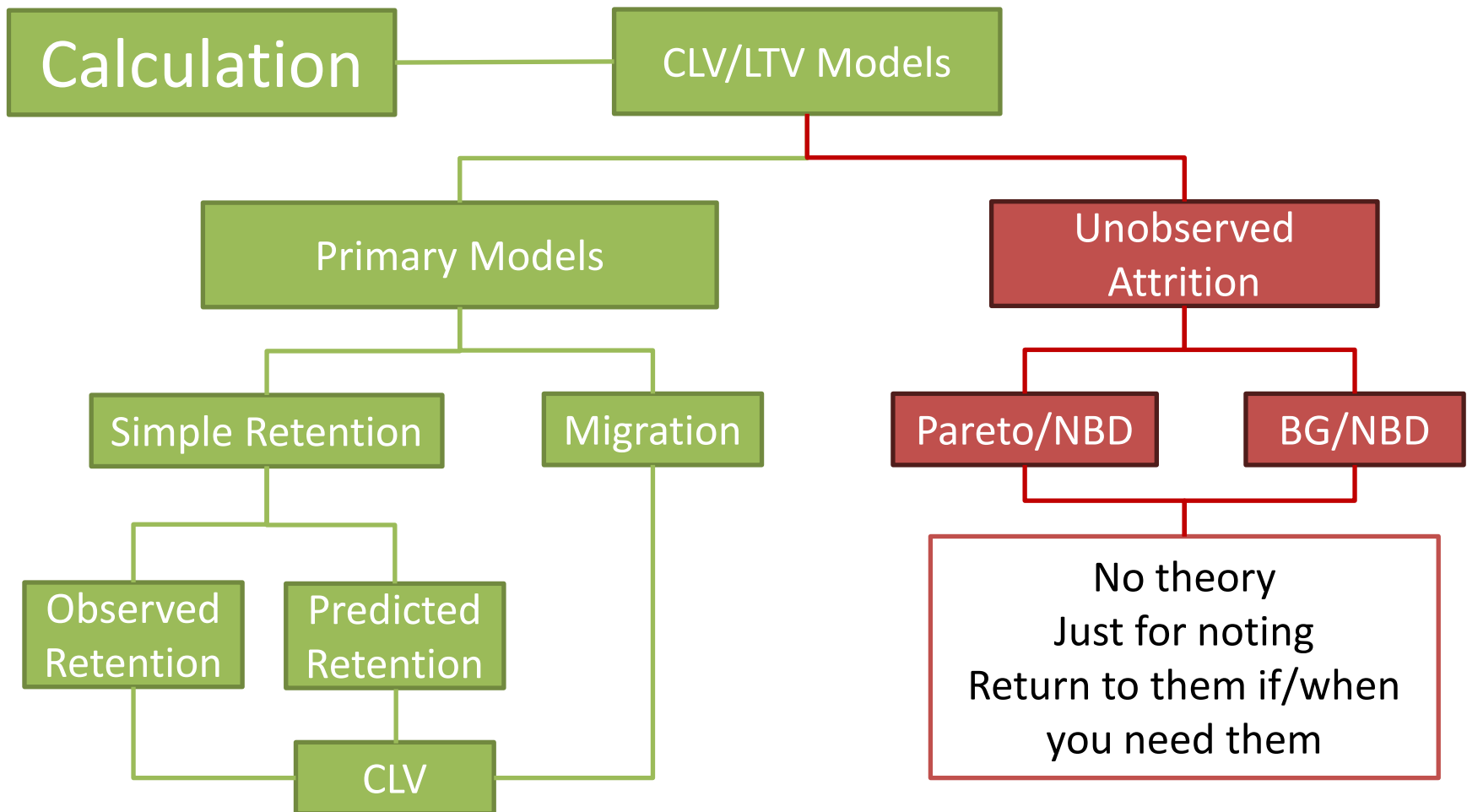
Calculation

CLV/LTV Models

Unobserved Attrition

- Three models developed to date:

	Method		
	Pareto/NBD (P/NBD)	Beta geometric/NBD (BG/NBD)	Beta geometric/beta binomial (BG/BB)
	Schmittlein et al. (1987)	Fader et al. (2005)	Fader et al. (2004)
Phenomena			
Customer alive	$\tau = \text{lifetime} \sim \text{Exp}(\mu)$	$p = \text{probability "die" after each purchase}$	$q = \text{probability "die" after each purchase}$
Heterogeneity	$\mu \sim \text{gamma}(s, \beta)$	$p \sim \text{Beta}(a, b)$	$q \sim \text{Beta}(\gamma, \delta)$
Purchase rate	$x = \text{number of purchases in time } t \sim \text{Poisson}(\lambda)$	$x = \text{number of purchases in time } t \sim \text{Poisson}(\lambda)$	$p = \text{probability of purchase in each period}$
Heterogeneity	$\lambda \sim \text{gamma}(r, \alpha)$	$\lambda \sim \text{gamma}(r, \alpha)$	$p \sim \text{Beta}(\alpha, \beta)$
Summary			
Lifetimes	Exponential	Geometric	Geometric
Purchases	Poisson	Poisson	Bernoulli
Interpurchase	Exponential	Exponential	Geometric



Homework



No more classwork!



Use any AI tool to generate a characterisation of yourself doing this course. Add your name and some details that tell me about yourself. Also, distinguish between NWU and UWC. Email the image to me when you're ready.

Same Gemini Pro model, two different chats

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